Real Images

Exam code: 0625 & 0972

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Written by: Katie M
Reviewed by: Caroline Carroll
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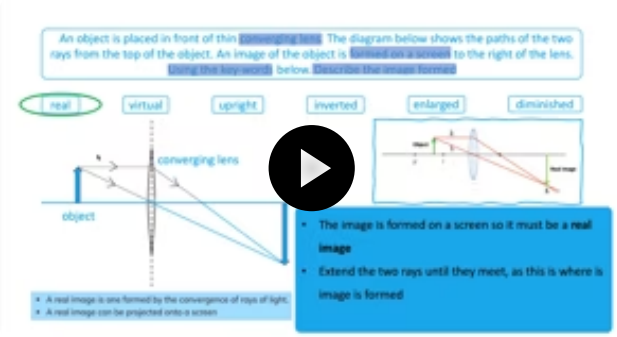


Did this video help you? Yes No

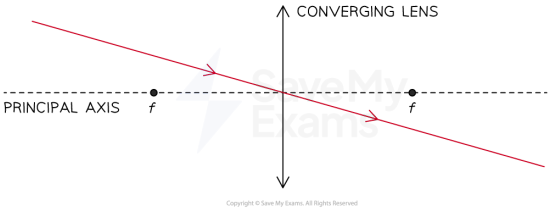
Help me understand

Converging lens – real image

Constructing converging ray diagrams

- The three main rules for constructing ray diagrams are as follows:

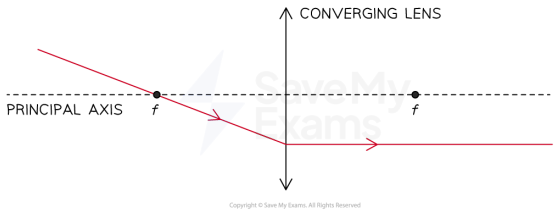
1. Rays passing through the principal axis will pass through the optical centre of the lens undeviated



2. Rays that are parallel to the principal axis will be refracted and pass through the focal point *f*

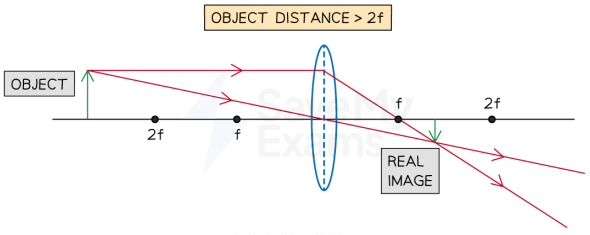
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

3. Rays passing through the focal point *f* will emerge parallel to the principal axis



Drawing converging ray diagrams of real images

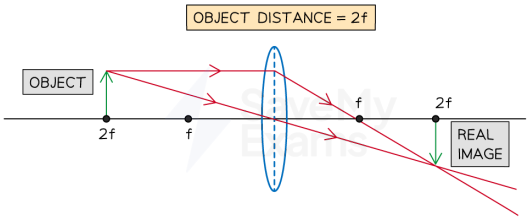
Object placed beyond 2f



- When an **object** is placed **beyond 2*f*** (to the left of the lens), the **image** that forms (to the right of the lens) will have the following properties:

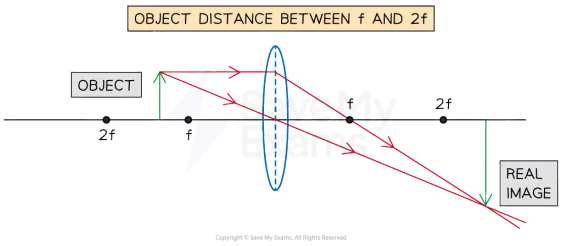
The image forms…	between *f* and 2*f*
The nature of the image is…	real
The orientation of the image is…	inverted
The size of the image is…	diminished

Object placed at 2f



The image forms...	at 2*f*
The nature of the image is...	real
The orientation of the image is...	inverted
The size of the image is...	the same

Object placed between f and 2f



- When an **object** is placed **between *f* and 2*f*** (to the left of the lens), the **image** that forms (to the right of the lens) will have the following properties:

The image forms...	beyond 2*f*
The nature of the image is...	real
The orientation of the image is...	inverted
The size of the image is...	magnified

Worked Example

Draw a ray diagram to show how a converging lens can be used to form a diminished image of a real object.

Label the object, image and principal foci of the lens on your diagram.

Answer:

Step 1: Start by drawing and labelling a principal axis and the lens as a line or a very thin ellipse

Step 2: Mark and label the focal points on each side of the lens

Step 3: Draw and label the object at a distance greater than the focal length on the left side of the lens

- Tip: For a diminished image the object should be placed a distance of at least 2F away from the lens

Step 4: Draw a ray through the optical centre of the lens

Step 5: Draw a second ray from the object to the lens which is parallel to the principal axis

Step 6: Draw the continuation of the ray passing through the focal point on the right side of the lens

Step 8: Check your final image and make sure everything is included to gain the marks

- For a three-mark question, examiners will be looking for:
  - One ray drawn through the optical centre of the lens
  - A second ray drawn which produces a diminished (smaller) image (which must pass through a labelled focal point)
  - Both the object and the image must be drawn and labelled correctly

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